

Chapter 16

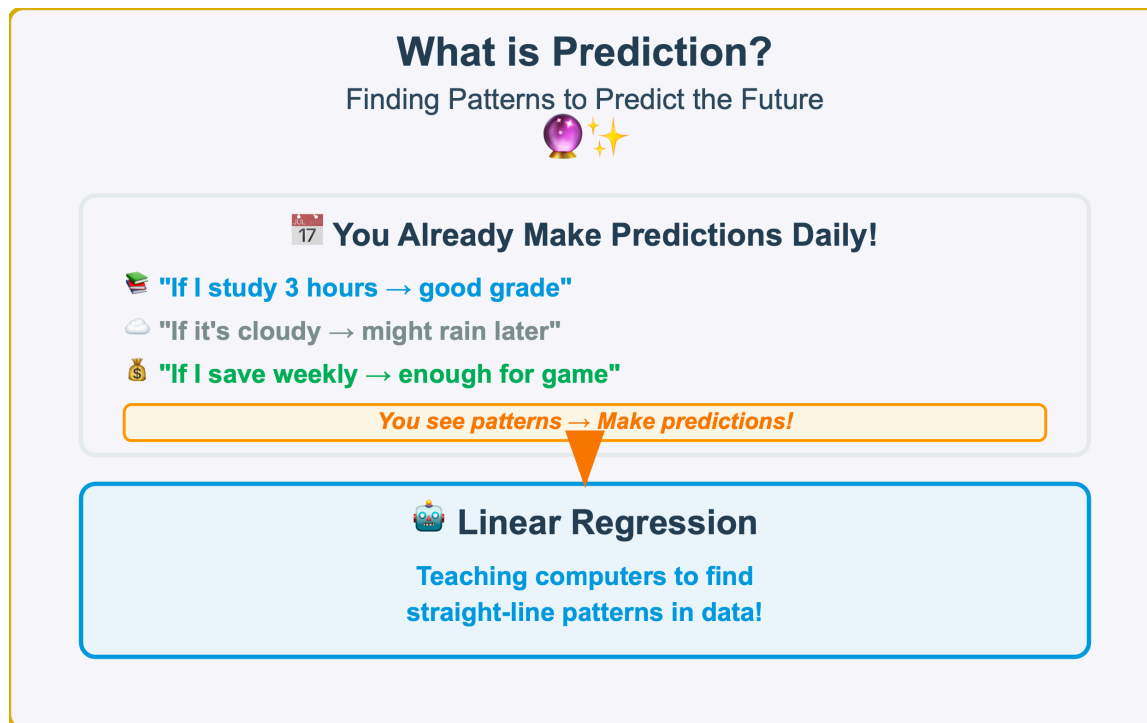


Figure 16.1 — What is Prediction? — Finding Patterns to Predict the Future. Students already make predictions daily based on patterns they observe. Linear regression teaches computers to find straight—line patterns in data to make similar predictions.

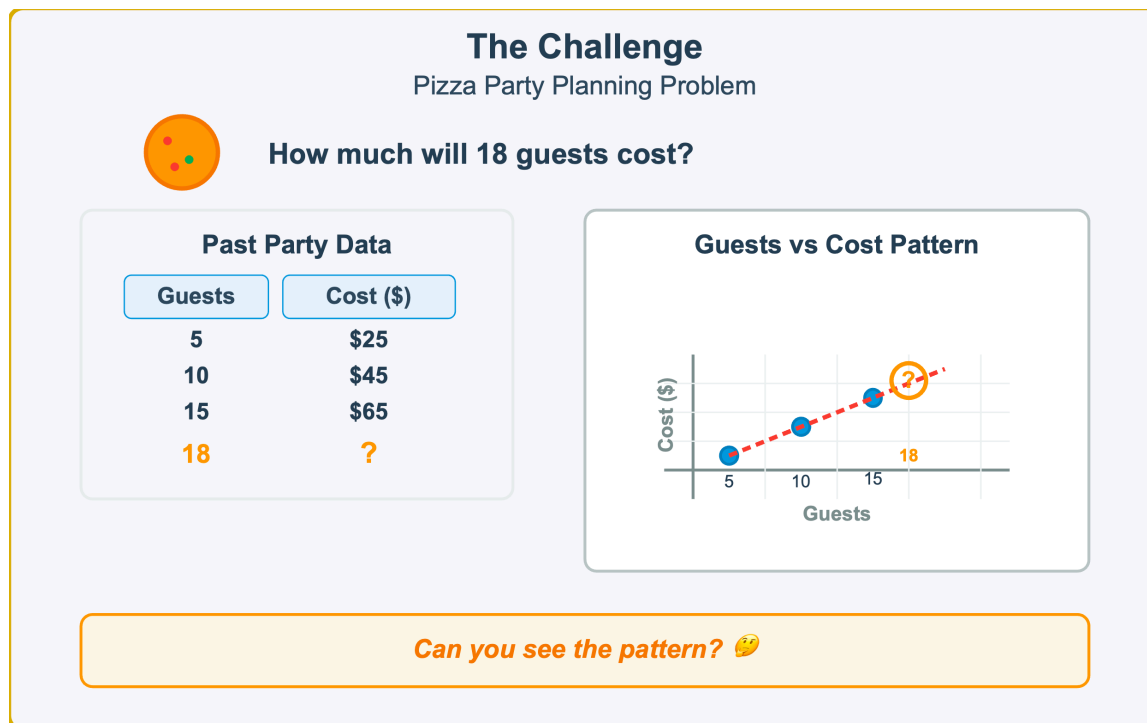


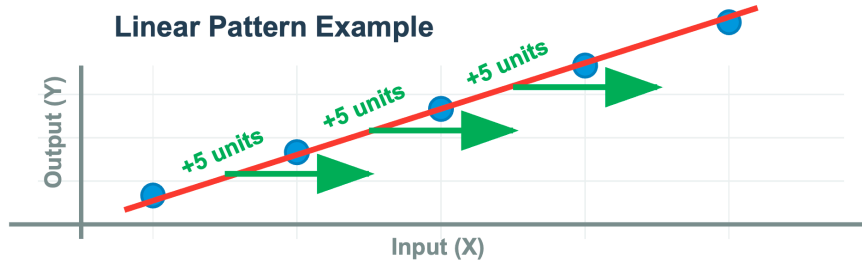
Figure 16.2 — The Challenge — Pizza Party Planning Problem. Students analyze past party data to discover the linear relationship between number of guests and total cost, then predict expenses for 18 guests using pattern recognition.

What Makes a Relationship "Linear"?

When one thing changes, the other changes
by a consistent amount

= Straight Line Pattern

Linear Pattern Example



🔑 Consistent Change = Linear Relationship

Figure 16.3 — What Makes a Relationship “Linear”? A linear relationship exists when one variable changes by a consistent amount as another variable increases, creating a straight line pattern in the data.

Finding the Linear Pattern

Machine Learning Discovers the Line

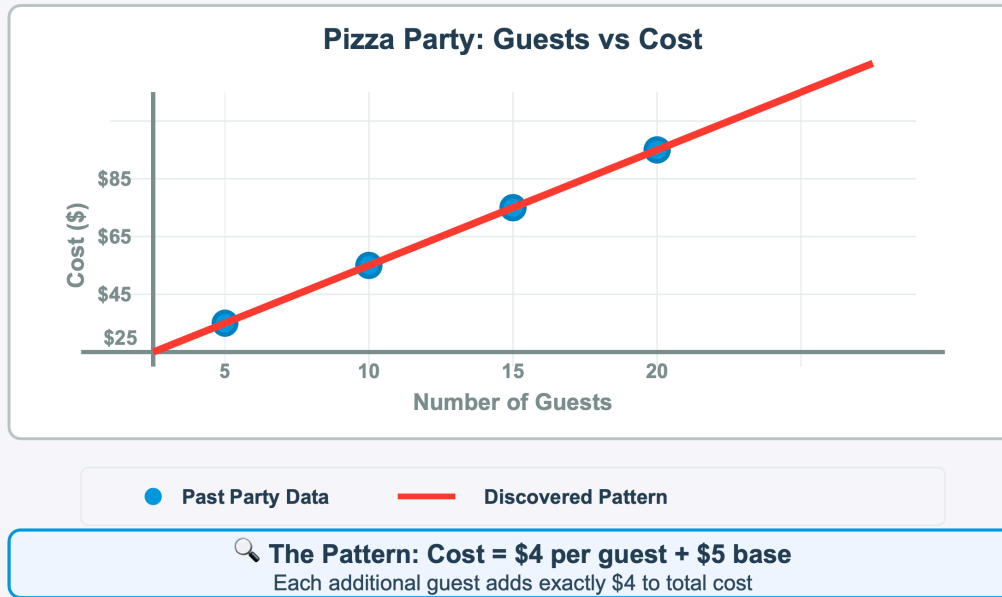


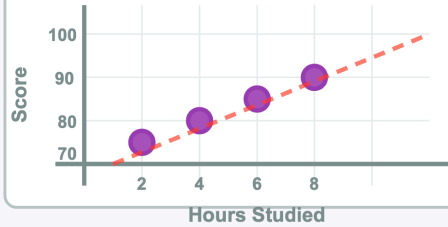
Figure 16.4 — Finding the Linear Pattern — Machine Learning Discovers the Line. By analyzing the pizza party data points, machine learning identifies the linear relationship: $\text{Cost} = \$4 \text{ per guest} + \5 base , where each additional guest adds exactly \$4 to the total cost.

Try It Yourself: Find the Pattern

Study Time vs Test Score Data

Hours Studied	Test Score
2	70
4	80
6	90

Study Hours vs Test Score



🎯 Your Challenge:

1. Do the points form a straight line?
2. What's the slope (change per hour)?
3. Find the equation! 🤔
4. Predict score for 8 hours study

💡 Hint: Look for consistent change between data points!

Figure 16.5 — Try It Yourself — Find the Pattern. Students analyze study time versus test score data to practice identifying linear relationships, determining slope, and finding the equation that describes the pattern.

What is a "Best-Fit" Line?

Real data is messy! The best-fit line comes closest to ALL the points (not perfect, but best overall)

Perfect Data ✨
All points on the line!



Real-World Data 🗑️
Points scattered around line



Realistic Pizza Party Data



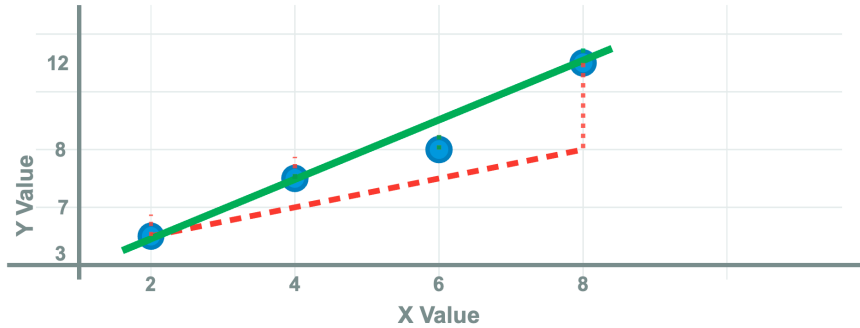
📊 The red line minimizes overall distance to ALL points

Figure 16.6 — What is a “Best—Fit” Line? Real data is messy and doesn’t form perfect lines. The best—fit line comes closest to all data points overall, minimizing the total distance from the line to every point in the dataset.

Visual Understanding of "Best Fit"

Which line is closer to ALL the points?

Comparing Different Lines to Data



● Data Points

- - - Poor Fit ($y=x$)

— Better Fit ($y=1.5x+0.5$)

👁️ **Green line = smaller total distance to all points**

Machine learning finds the line with smallest total distance

Figure 16.7 — Visual Understanding of “Best Fit”. Comparing two different lines through the same data points shows that the green line ($y=1.5x+0.5$) has smaller total distance to all points than the red line ($y=x$), demonstrating how machine learning finds the optimal fit.

Becoming a Human Regression Algorithm

Calculate slope and intercept by hand!

Exercise → Happiness Data

Hours	Happy
1	3
3	6
6	9

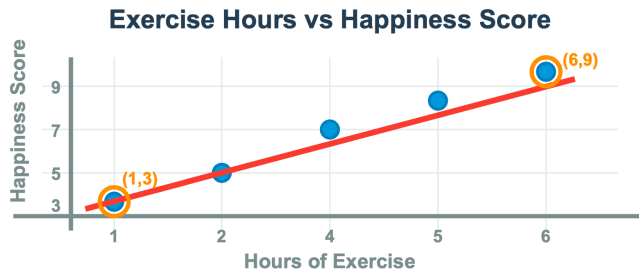
Manual Calculation

Step 1: Pick two points

First: (1,3) Last: (6,9)

Step 2: Calculate slope

Slope = $(9-3)/(6-1) = 6/5 = 1.2$



Final Equation:

$$y = 1.2x + 1.8$$

Each hour adds
1.2 happiness points!

🎉 You just did linear regression by hand! This is exactly what computers do automatically.

Figure 16.8 — Becoming a Human Regression Algorithm. Students learn to calculate slope and intercept manually using two data points from exercise vs. happiness data, discovering the linear relationship $y = 1.2x + 1.8$ where each hour of exercise adds 1.2 happiness points.

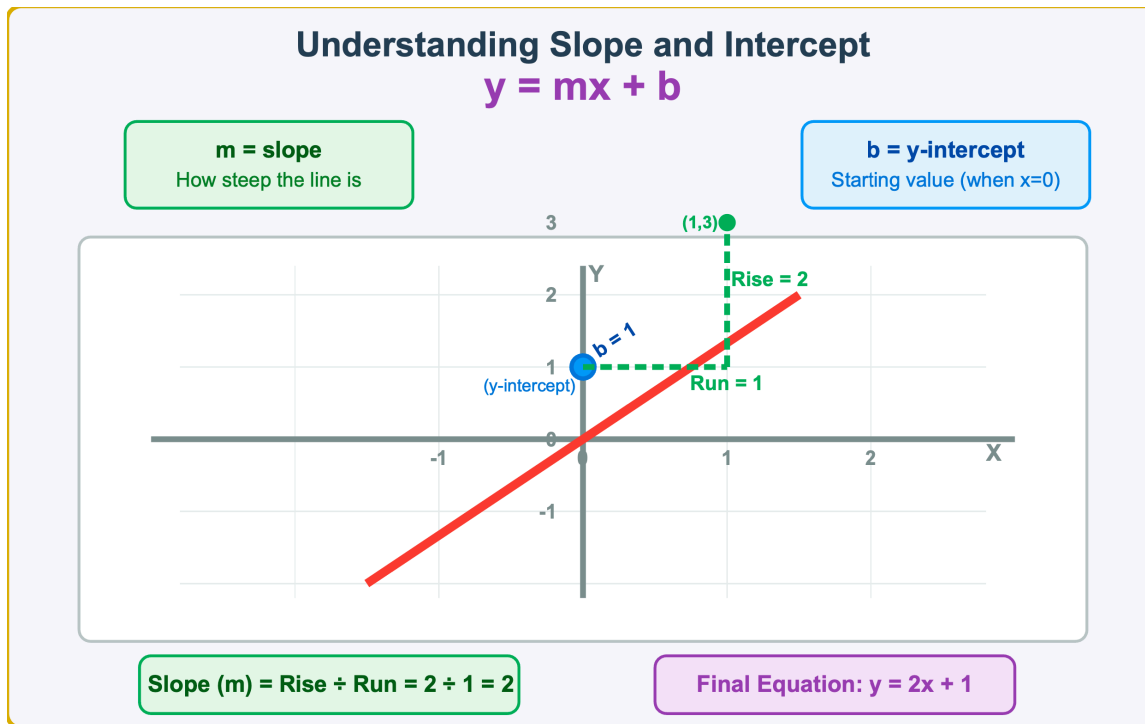


Figure 16.9 — Understanding Slope and Intercept. The linear equation $y = mx + b$ consists of slope ($m = \text{rise/run} = \text{steepness}$) and y—intercept ($b = \text{starting value when } x = 0$). This example shows $y = 2x + 1$, where slope = 2 and y—intercept = 1.

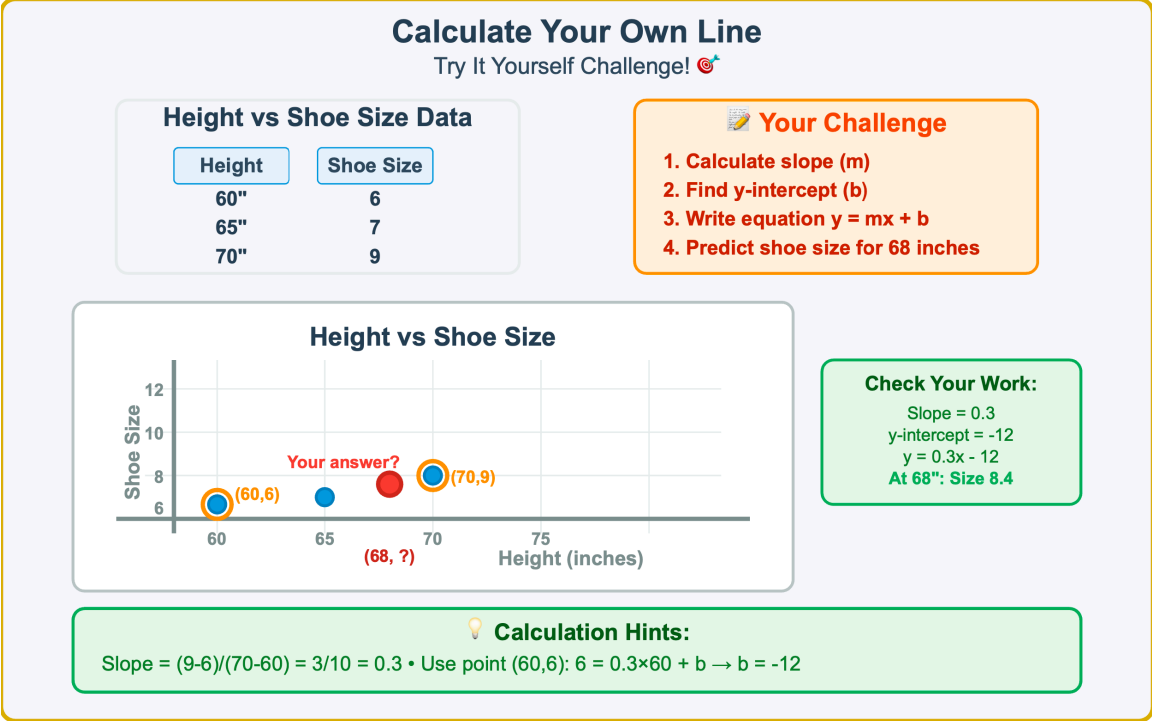


Figure 16.10 — Calculate Your Own Line — Try It Yourself Challenge. Students practice linear regression calculations using height vs shoe size data, finding slope (0.33), y—intercept (—14), and predicting shoe size for a 68—inch height.

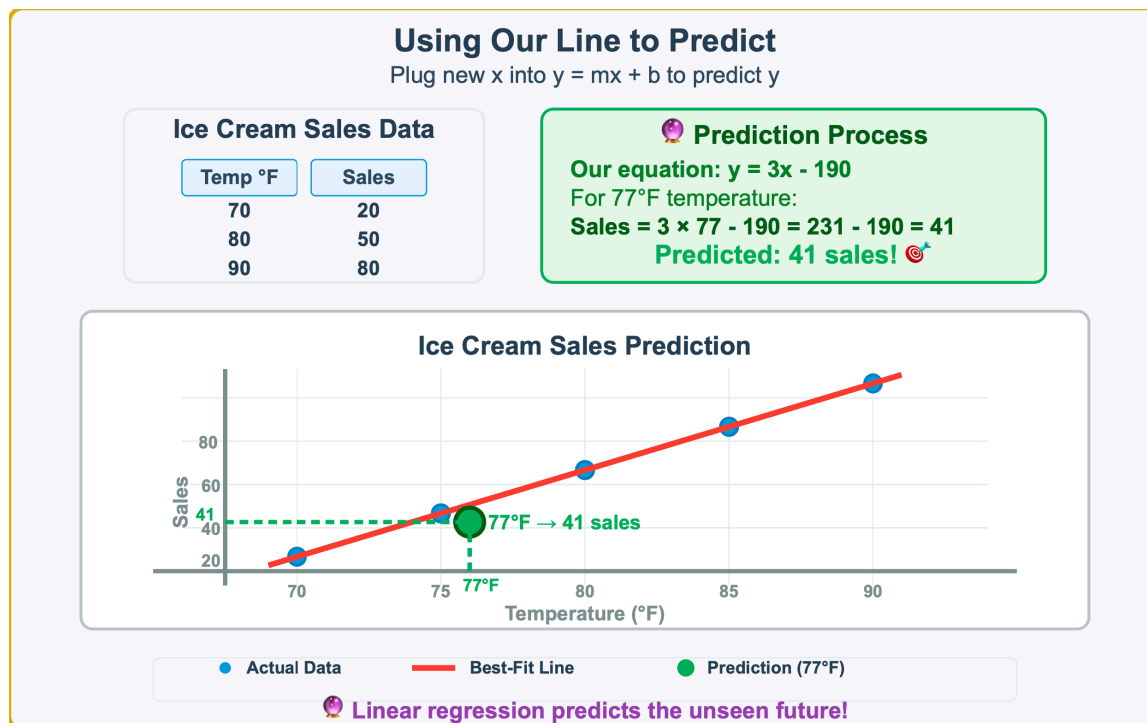


Figure 16.11 — Using Our Line to Predict. Once we have a linear equation ($y = 3x - 190$), we can predict unknown values by substituting new x values. Here, predicting ice cream sales at 77°F gives us 41 sales, demonstrating how linear regression predicts the unseen future.

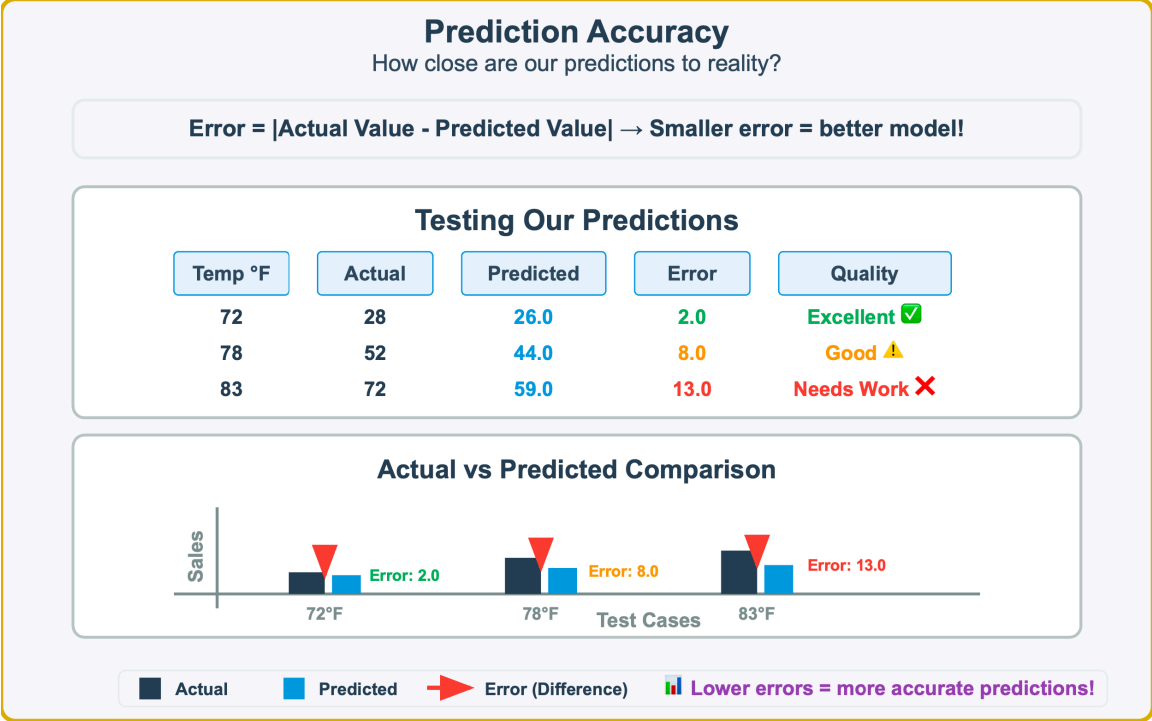


Figure 16.12 — Prediction Accuracy. Model quality is measured by comparing actual values to predicted values, calculating error as the absolute difference. Smaller errors indicate more accurate predictions, as shown in this ice cream sales validation test.

Make Your Own Predictions

Try It Yourself Challenge! 🎯

Study Hours → Test Score

Hours	Score
1	65
3	75
5	85

Your Tasks

1. Find slope & intercept
2. Write equation
3. Predict for 3.5 hours (interpolation)
4. Predict for 7 hours (extrapolation)

Study Hours vs Test Score



Key Concepts:

Interpolation 📍
Predicting within data range
(More reliable)

Extrapolation ⚠️
Predicting beyond data range
(Less reliable)

Calculation Hints:

Slope = 5 points per hour • Equation: $\text{Score} = 5 \times \text{Hours} + 60$
 $3.5h \rightarrow 77.5$ points (reliable) • $7h \rightarrow 95$ points (extrapolation - less trustworthy)

Figure 16.13 — Make Your Own Predictions — Try It Yourself Challenge. Students practice making predictions using study hours vs test score data, learning the difference between interpolation (3.5 hours = reliable) and extrapolation (7 hours = less reliable) when predicting beyond the data range.

$$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

Formula for computing the mean

$$b = \bar{y} - m \cdot \bar{x}$$

Formula for calculate the intercept

Understanding Scipy Linregress & OLS

What the computer is actually doing

What is `scipy.linregress`?

A computer tool that automatically finds the best-fit line
It does all the math we did by hand, but much faster!

What is Ordinary Least Squares (OLS)?

The mathematical method to find the line that minimizes errors
"Least Squares" = smallest total of squared distances

Why "Least Squares"?



The OLS Formulas

Slope (m):

$$m = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

Intercept (b):

$$b = \bar{y} - m\bar{x}$$

$\bar{x}$ = average of x values

$\bar{y}$ = average of y values

What `scipy.linregress` does automatically:

1. Calculates averages ($\bar{x}$, $\bar{y}$)
2. Applies OLS formulas
3. Returns slope & intercept

Same result as our manual method, but handles any amount of data instantly!

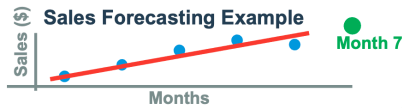
Figure 16.14 — Understanding Scipy Linregress & OLS. The `scipy.linregress` function automates the Ordinary Least Squares (OLS) method, which finds the best—fit line by minimizing the sum of squared errors. It performs the same calculations students did manually but can handle any amount of data instantly.

Linear Regression in the Real World

It's everywhere! Here are some examples:



Business & Economics



Examples:

- Forecast monthly sales
- Predict stock prices
- Estimate production costs



Science & Medicine

Examples:

- Drug dosage vs patient weight
- Temperature vs plant growth
- Study hours vs test performance

Dosage vs Weight



Sports & Entertainment

Examples:

- Training hours vs performance
- Movie budget vs box office
- Practice time vs game scores

Training vs Performance



Linear regression helps us understand relationships and make predictions in almost every field!

From business forecasting to medical research to sports analytics

Figure 16.15 — Linear Regression in the Real World. Linear regression is used across many fields including business (sales forecasting, budget planning), science and medicine (drug dosage, plant growth studies), and sports and entertainment (training performance, box office predictions), helping professionals understand relationships and make data—driven predictions.

Choose Your Application

Pick a scenario for your regression analysis!



1. Sports

Training hours vs performance improvement
Example: Basketball practice hours → free throw %
Perfect for sports enthusiasts and athletes



2. Environment

Temperature vs energy consumption
Example: Daily temperature → kilowatt-hours used
Great for environmental science students



3. Education

Class size vs student performance
Example: Number of students → average test score
Ideal for future teachers and education majors



4. Technology

RAM amount vs computer speed
Example: Gigabytes of RAM → benchmark score
Perfect for computer science and tech students



Pick one scenario that interests you most!

Then follow the tasks on the next figure to complete your analysis
Choose based on your interests - all scenarios teach the same concepts!

Figure 16.16 — Application Scenario Selection for Regression Analysis. Students choose from four real—world contexts to apply regression analysis concepts, making the learning personally relevant and engaging.

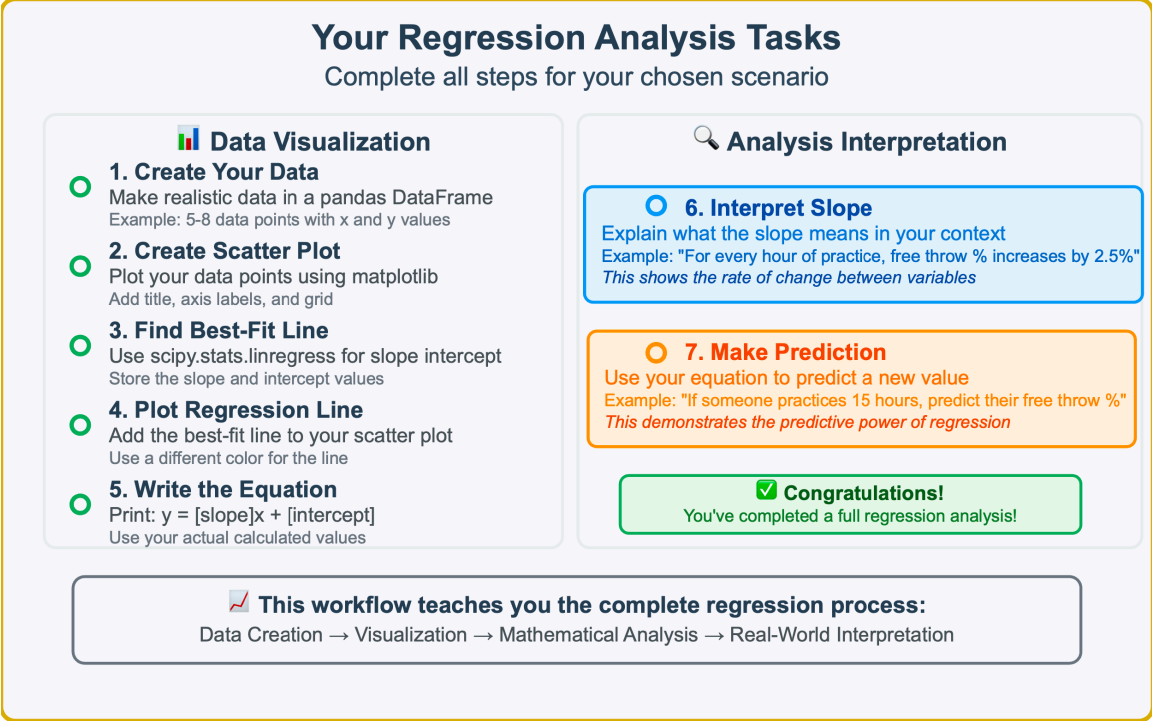


Figure 16.17 — Regression Analysis Task Checklist. Students follow this seven—step workflow to complete a full regression analysis on their chosen scenario from Figure 16.16, building practical programming and statistical analysis skills.

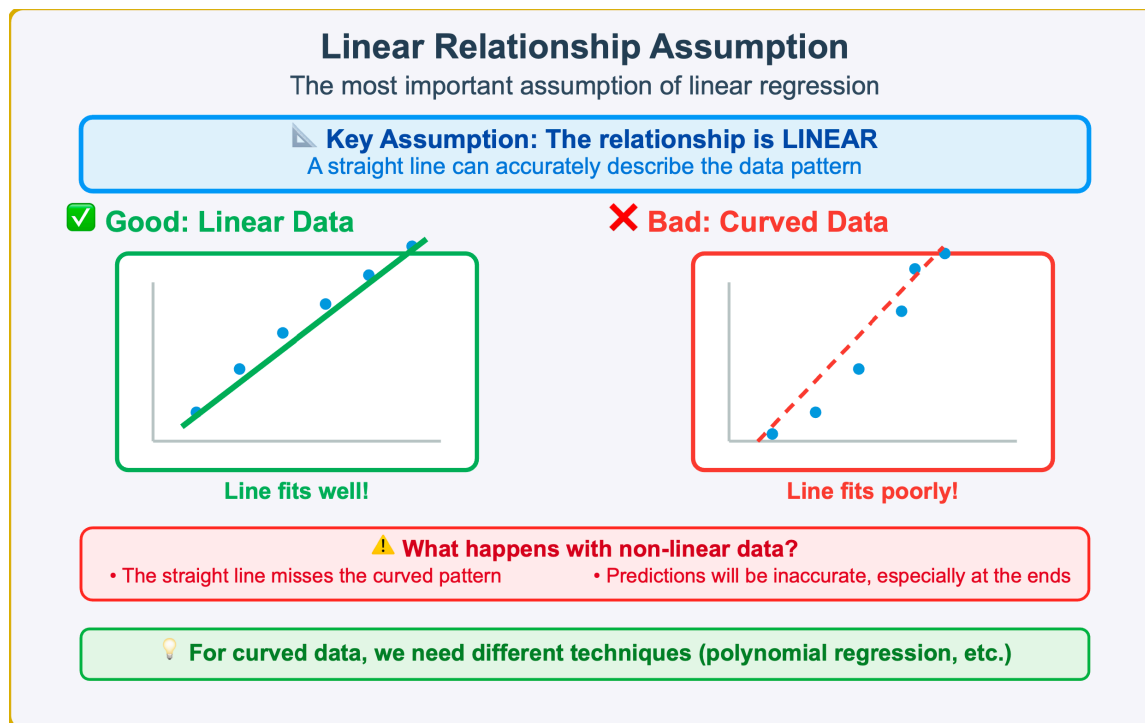


Figure 16.18 — Linear Relationship Assumption in Regression Analysis. The fundamental assumption of linear regression is that data follows a straight—line pattern. When this assumption is violated (curved data), linear regression produces poor fits and inaccurate predictions, requiring alternative techniques like polynomial regression.

Correlation $\neq$ Causation

Linear regression finds relationships, not causes

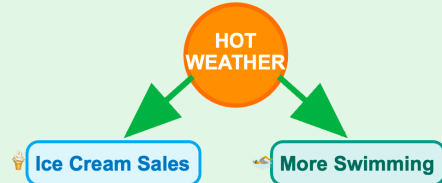
⚠️ Common Mistake

Example: Ice cream sales vs drowning incidents

- Both increase during summer months
- Linear regression shows strong correlation
- Data appears to suggest a direct relationship

❌ **WRONG: Ice cream causes drowning!**

✓ Correct Understanding



🤔 Always think: What could be the common cause?

Linear regression reveals patterns, but human reasoning determines causes

Look for confounding variables that might explain both observations

💡 Remember: Strong correlation can exist without any causal relationship

Use regression for prediction and pattern discovery, not for proving cause and effect

Figure 16.19 — Correlation vs. Causation in Regression Analysis. The classic ice cream and drowning example illustrates how linear regression can reveal strong correlations that have no causal relationship. Hot weather is the common cause that increases both ice cream sales and swimming (leading to more drownings), demonstrating the importance of looking for confounding variables when interpreting regression results.

Complete Regression Projects

Real-world challenges to practice your skills!

Challenge 1: School Performance Analysis

Student Performance Data	
Study Hours/Week	Final Grade
5	72
8	78
12	88
15	92

Your Mission:

- Analyze the relationship between study time and grades
- Create visualizations and mathematical models
- Make predictions and interpret results
- Assess the reliability of your analysis

Your Comprehensive Analysis:

- Create Scatter Plot**
Plot study hours vs final grades with proper labels
- Calculate Best-Fit Line**
Use `scipy.stats.linregress` to get slope & intercept
- Draw Regression Line**
Add the best-fit line to your scatter plot
- Write Equation**
Print: $\text{grade} = [\text{slope}] \times \text{hours} + [\text{intercept}]$
- Interpret Slope**
What does each extra study hour add to the grade?
- Make Prediction**
Predict grade for 14 hours of study and plot it
- Assess Reliability**
How well do points follow the line? Good fit?
- Draw Conclusions**
What insights can help students improve?

Ready for More Challenges?

Weather Analysis • Sports Performance • Business Sales • Environmental Data

Apply the same 8-step process to any dataset that interests you!

Figure 16.20 — Complete Regression Projects. Students apply all regression analysis skills in a comprehensive school performance study. The 8—step process guides learners through data visualization, mathematical modeling, interpretation, and conclusion drawing, providing a complete methodology for any regression analysis project.

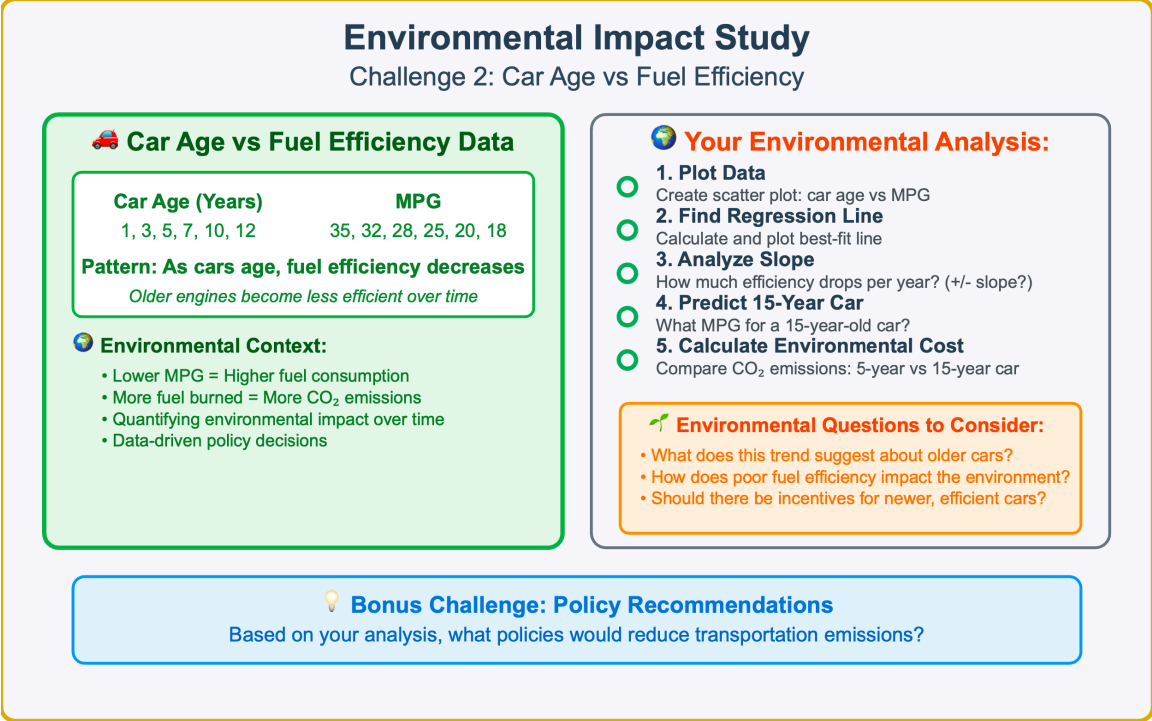


Figure 16.21 — Environmental Impact Study. Students apply regression analysis to examine how car age affects fuel efficiency, connecting mathematical modeling to environmental science. The analysis reveals quantitative relationships between vehicle age and emissions, demonstrating how data science informs environmental policy decisions.

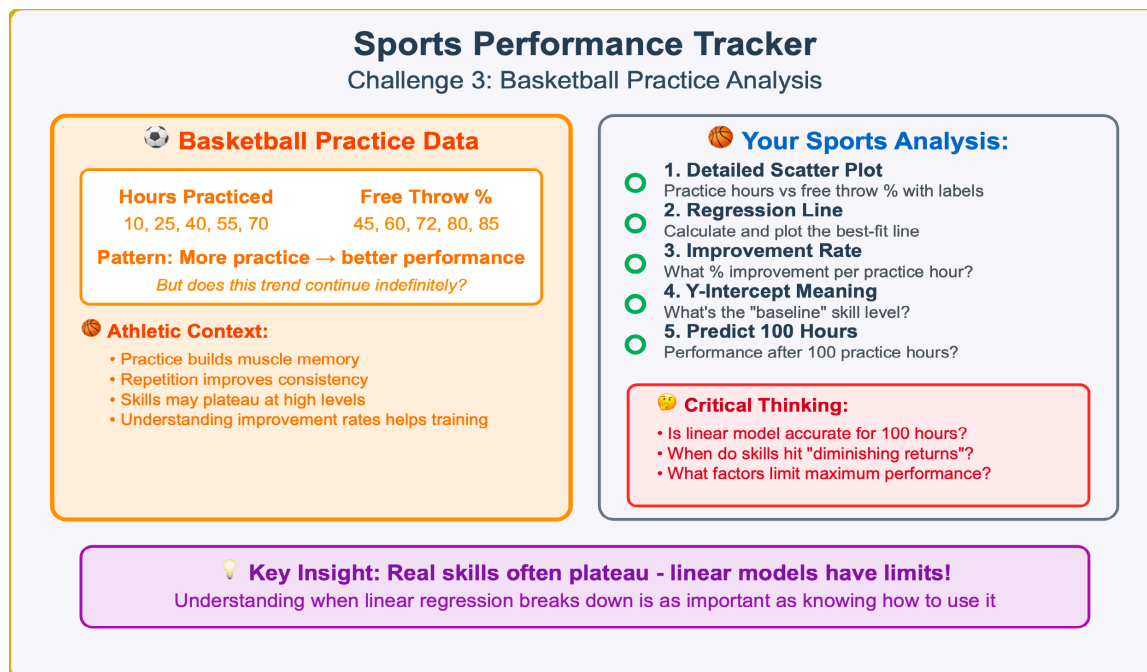


Figure 16.22 — Sports Performance Tracker. Students analyze basketball practice data using regression analysis while learning about model limitations. The analysis reveals how practice improves performance but also teaches critical thinking about when linear models may not accurately represent real—world complexity, particularly the concept of skill plateaus and diminishing returns.